# Ansh Rathod

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## EDUCATION

**Master of Science in Information Systems** Sep 2023 – Dec 2025
*Northeastern University, Boston, MA*
**Relevant Coursework:** Python & Analytics Technology, Business Processes, Predictive Analytics, Probability Theory & Statistics

**Bachelor of Technology in Computer Science** Jul 2018 – Jun 2022
*National Institute of Technology Nagpur, India*
**Relevant Coursework:** Natural Language Processing, Machine Learning in Python, Big Data, Data Structures & Algorithms

## PROFESSIONAL EXPERIENCE

**AI/ML Engineer** Aug 2025 – Mar 2026
*Baxter | Deerfield, IL*

- Built a cloud data pipeline using Python, AWS Glue, PySpark, and EC2, cutting execution time from 45 to 10 minutes, and developed a Power BI star schema dashboard reducing report generation by 30%.
- Analyzed 5 years of claims data with Python (SciPy, Pandas), t-tests, and Chi-squared tests to reduce readmission rates by 15%; optimized SQL queries for audit and risk detection, exploring XGBoost and NLP models for predictive insights.

**Data Analyst** Sep 2024 – Feb 2025
*Northeastern University | Boston, MA*

- Optimized SQL pipelines in Snowflake to process 15,000+ student applications, cutting query runtime by **30%** and delivering timely admissions insights.
- Automated data categorization using SQL window functions and CTEs; built Power BI dashboards providing real-time applicant tracking for the admissions team.

**Software Developer** Jun 2022 – Jul 2023
*iSoftwareFactory Services LLC | Mumbai, India*

- Architected and deployed a full-stack Q&A platform using MongoDB, Express.js, React.js, and Node.js (MERN) with scalable RESTful APIs, reducing production response times.
- Integrated Razorpay payment gateway for subscription-based features; shipped releases on schedule using Agile/Scrum and Git-based version control.

**Data Science/Business Analytics Intern** Jul 2021 – Dec 2021
*Ernst & Young | Mumbai, India*

- Developed and deployed a robust XGBoost model with dimensionality reduction for high-cardinality features, achieving 94.9% cross-validation accuracy while reducing computation cost and improving generalization.
- Implemented clustering-based signer identification workflows using DBSCAN, hierarchical clustering, and threshold-based clustering, with automated parameter optimization for improved cross-document robustness.

## PROJECTS

**Generative AI Supply Chain Analytics Tool** Sep 2024 – Dec 2024

- Built a Retrieval-Augmented Generation (RAG) system using OpenAI GPT-4, Pinecone, and Streamlit for semantic search and supply chain decision-making, improving forecasting precision by **25%**.
- Automated data collection and preprocessing using BeautifulSoup, Selenium, Pandas, and NumPy; engineered a flexible pipeline supporting weather data integration.

**RideFlow — Real-Time Data Engineering Pipeline (**Uber-Inspired**)** Dec 2024 – Mar 2025

- Built a scalable end-to-end real-time data pipeline using **Apache Spark (PySpark)** and **Azure Event Hub / Kafka** to ingest, clean, and process ride events, improving data throughput and pipeline efficiency by 40%.
- Designed a **Star Schema data warehouse** with optimized fact and dimension tables using **Spark Structured Streaming** for real-time transformations and aggregations, reducing processing latency by 25%.

## TECHNICAL SKILLS

**Programming Languages:** Python, R, SQL, MATLAB, JavaScript, Java, HTML/CSS, C

**Database & Cloud Tools:** MySQL, AWS (S3, EC2, Bedrock, Glue), MS Azure, Git, React, Docker, GCP, Pinecone, Apache Spark, Hadoop, Streamlit, Redshift, Snowflake, Talend, Alteryx, NoSQL, Power Query, Power Automate, Azure Data Factory

**Deep Learning:** NLP Models, LangChain, OpenAI, PyTorch, TensorFlow, scikit-learn, Hugging Face, Pandas, Keras

**Tools & Technologies:** Excel (VBA, VLOOKUP), Tableau, Power BI, Oracle SQL, MongoDB, Google Analytics, Adobe Analytics, t-test, Alteryx, Shiny, Streamlit, DBT, Data Warehousing, Data Mining, PowerPoint, MS Office, Looker

**AI & Generative AI:** LangChain, OpenAI API, Claude (Anthropic), Hugging Face, Retrieval-Augmented Generation (RAG), LLM Fine-Tuning, Prompt Engineering, AI Agents, Vector Databases (Pinecone), AWS Bedrock, Multimodal AI